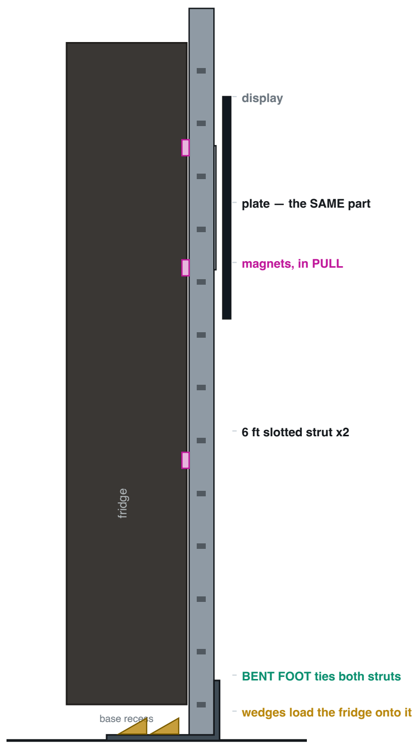
CONCEPT — superseding the hook, pending the fridge base

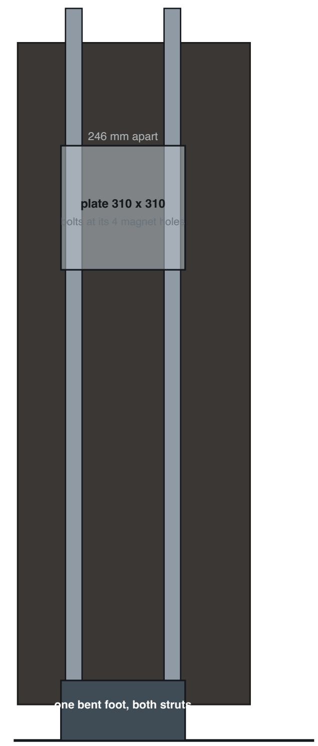
TWO STRUTS ON A BENT FOOT

A pair of 6 ft slotted struts up the side panel, tied at the base by one bent foot that slides under the fridge and is wedged so the appliance's own weight holds it down.

SIDE — the load path



FRONT — looking at the side panel



THE NUMBERS

Strength, foot 29.6 N·m over a 310 mm foot	SF 10
Strength, strut 41 MPa in the pull direction	SF 6.1
Torsion, edge press 812 lb·mm over 246 mm spacing	3.3 lb per strut
Magnets, propping 5 lb of PULL shared over 4	1.25 lb each
Sway if UNPROPPED 4.7 mm at the screen	magnets fix this, not the foot

THE WEDGE — what it actually has to do

wedge 100 mm inboard -> 66.6 lb on the foot, 29% of the appliance
wedge 150 mm inboard -> 44.4 lb on the foot, 19% of the appliance
wedge 200 mm inboard -> 33.3 lb on the foot, 15% of the appliance
wedge 250 mm inboard -> 26.6 lb on the foot, 12% of the appliance

WHAT IS SETTLED AND WHAT IS NOT

SETTLED by geometry

Strut orientation is forced: flat back on the panel, slots out. That is also the good one — a wide flat magnet bearing.

The plate bolts on UNMODIFIED. Its four O8.5 magnet holes are 246 mm apart, which is the strut spacing, and O8.5 clears a 5/16 channel bolt.

The foot is the same process as the hook: laser plus ONE 90 degree bend, same material, same supplier.

OPEN — needs the fridge

How high the cabinet base sits off the floor: that is the foot plus wedge budget.

Whether wedging a fifth of the appliance onto the foot rocks it or dents the base pan. Wedging is uncontrolled by nature — how hard someone drove it is not a designable quantity.

So the wedge is BACKUP. The magnets carry the case on their own, in pull, at 49x.